

TURN 7 CONTINUATION: SINGULAR-FIBRE ELIMINATION AND SMALL-SHAPE RIGIDITY IN THE ORIGINAL ERDŐS–STRAUS SOURCE

THE CLANKERS

ABSTRACT. An original exterior state determines a marked integral cubic with coordinates $X = a^2/u$, $Y = 2a$ and defects $\alpha = X - R$, $\beta = X - D$. The original nonresidue grade divides $\alpha\beta - 1$. We construct a same-prime middle return from every diagonal cofactor state, and prove that every hard-prime exterior state with both defects of absolute value at most seven is on the singular family already responsible for the preceding cutoff's sharpness. It has an explicit middle state at its unchanged shell. Radius eight reduces to one specified quartic. All source maps, image conditions, fibres and channel normalizations are retained. Universal ES existence remains unproved.

1. THE UNRESOLVED STEP AND THE BRANCH REMOVED HERE

The universal assertion remains that every prescribed hard prime supplies an original positive exterior or middle state. The complete atlases from the preceding work decide a fixed input; their finite index bounds do not prove occupancy. We do not advance to a universally successful algorithm under an assumed occupancy statement.

The previous exterior cutoff has an asymptotically sharp integer family. Here we determine its exact marked cubic geometry, prove that its diagonal cofactor locus already has an original middle return, and classify every hard-prime exterior state whose two shape defects have absolute value at most seven. Every such state is on that same singular family, and has two original middle orientations at its *unchanged* distinguished denominator. At radius eight there is at most one nonsingular shape; its grade congruence also gives an explicit middle return, independently of solving its associated quartic. This is an existence-preserving branch reduction, not a proof of the remaining unbounded positive coefficient.

The original Type-I/II parametrizations are antecedents [1, §2]. The integer family itself occurs in the preceding exterior proof [3, §5]. No historical-priority determination is made for the present deductions. All algebra and the finite modular classification used here are proved below; no theorem about integral points on elliptic curves is imported.

2. ORIGINAL SOURCE AND THE MARKED CUBIC

Let $p \equiv 1 \pmod{8}$ be prime. An original exterior state has

$$p/4 < a < p/2, \quad R = 4a - p, \quad u \mid a^2, \quad R \mid 4u + 1.$$

Retain $a = \prod_l l^{e_l}$, $\beta_l = v_l(u) - e_l$, and the ordered normalization

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g), \quad \kappa = (pr + s)/R.$$

Then $a = hrs$, $u = hr^2$, $\gcd(r, s) = 1$, and $r < \kappa$. The ordered return is

$$(x, y, z) = (a, hsk, phr\kappa), \quad \frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \tag{1}$$

Its ordered inverse is $u = a^2/(Ry - pa)$. Prime valuations prove the integrality of the normalization. Multiplication proves (1), without identifying its ordered denominators with another channel's order.

Keep the original complement and cofactor

$$v = a^2/u = hs^2, \quad D = (4u + 1)/R = (r + \kappa)/s.$$

They satisfy

$$RD = 4u + 1, \quad pD + 1 = 4hr\kappa, \quad (p + R)^2 + 4v = 4vRD, \quad R, D \equiv 3 \pmod{4}. \quad (2)$$

The case $h = 1$ would make -1 a square modulo $R \equiv 3 \pmod{4}$. Thus $h > 1$. Also $\gcd(h, R) = \gcd(h, D) = 1$, and consequently $v \neq R, D$. No assertion $\gcd(v, D) = 1$ is made.

Define the *marked shape* and its coefficient

$$\alpha = v - R, \quad \beta = v - D, \quad B = \alpha\beta - 1, \quad X = v, \quad Y = 2a. \quad (3)$$

(The symbol β in (3) is not an original exponent β_l .) The pairs (α, β) and (R, D) are ordered data: interchanging them leaves the unmarked cubic equation unchanged but changes the reconstructed distinguished residual and, in general, the reconstructed prime.

Theorem 2.1 (Exact ordered shape curve and inverse). *Every original state maps to an integral point*

$$\boxed{Y^2 = X^3 - (\alpha + \beta)X^2 + BX.} \quad (4)$$

The complete inverse is

$$R = X - \alpha, \quad D = X - \beta, \quad a = Y/2, \quad p = 2Y - R, \quad u = (RD - 1)/4. \quad (5)$$

Its domain consists exactly of points of (4) with $X > 0$, positive even Y , positive $R, D \equiv 3 \pmod{4}$, $Y > R$, and the prime condition on the reconstructed $p \equiv 1 \pmod{8}$. The original normalization then reconstructs every marking. Moreover

$$\boxed{h \mid B, \quad p \equiv \alpha \pmod{h}, \quad p\beta \equiv 1 \pmod{h}.} \quad (6)$$

The Weierstrass discriminant of (4) is

$$\boxed{16B^2((\alpha - \beta)^2 + 4).} \quad (7)$$

Thus its only singular integer shapes are $(\alpha, \beta) = (1, 1)$ and $(-1, -1)$.

Proof. From $a^2 = uv$ and $4u = RD - 1$, $Y^2 = v(RD - 1) = v((v - \alpha)(v - \beta) - 1)$, proving (4). Conversely (4) gives $a^2 = X(RD - 1)/4 = Xu$, so u is a positive divisor of a^2 . The residue requirements make u integral. Equation (5) gives $R = 4a - p > 0$ and $2a < p$ precisely from $Y > R$. The original gate follows from $4u + 1 = RD$. The prime condition and $a < p$ imply the original unit condition. All formulas are literal inverses.

Reducing $RD - 1 = 4u$ modulo h gives $\alpha\beta - 1 \equiv 0$; reducing $p = 4hrs - R$ and $pD + 1 = 4hr\kappa$ gives the other two identities in (6). For the cubic $X(X^2 - (\alpha + \beta)X + B)$, the polynomial discriminant is $B^2((\alpha + \beta)^2 - 4B)$. The Weierstrass discriminant is sixteen times the polynomial discriminant, giving (7). Its second factor is $(\alpha - \beta)^2 + 4 > 0$. Thus singularity means $\alpha\beta = 1$, whose integer possibilities are the two stated shapes. $\square$

For a fixed original p and shape, the point must also retain the line

$$\boxed{2Y = p + X - \alpha.} \quad (8)$$

Eliminating Y yields the explicit cubic

$$4X^3 - [4(\alpha + \beta) + 1]X^2 + [4B - 2(p - \alpha)]X - (p - \alpha)^2 = 0. \quad (9)$$

There are at most three original states at a fixed (p, α, β) . This is a fibre bound, not an existence assertion. The integer coefficient pushforward that forgets X, Y sums those actual source coordinates; its kernel is generated by their same-shape differences. Retaining X, Y gives the bijection of Theorem 2.1. The point $(0, 0)$ of a nonsingular curve is outside the original positive source.

3. ORIGINAL CHARACTER CONSTRAINTS

Here and below, a *hard prime* means

$$p \bmod 840 \in \{1, 121, 169, 289, 361, 529\}.$$

Lemma 3.1. *For every original square divisor, $(u/R) = (u/p)$, where the first symbol is Jacobi. In particular $(h/p) = -1$. If p is hard, every prime factor among 2, 3, 5, 7 has character +1, and h contains a prime factor greater than seven. Neither α nor β can be a positive integer square.*

Proof. For an odd prime $l \mid a$, reciprocity and $R \equiv -p \pmod{l}$ give $(l/R) = (-1/l)(-p/l) = (p/l) = (l/p)$. For 2 use $R \equiv -p \pmod{8}$ when $2 \mid a$. Multiply over the original exponents. The exterior gate has $u \equiv -1/4 \pmod{R}$ and $R \equiv 3 \pmod{4}$, so $(u/p) = -1$; $u = hr^2$ gives the assertion for h . For the six hard classes, $p \equiv 1 \pmod{24}$, $p \equiv 1, 4 \pmod{5}$, and $p \equiv 1, 2, 4 \pmod{7}$, proving the four residue assertions. These are the ordinary reciprocity and supplementary laws [2, §6.4].

If $\alpha = c^2 > 0$, then every odd $l \mid h$ sees $p \equiv c^2 \pmod{l}$ by (6). The value is a nonzero square because $p \nmid h$. Thus $(l/p) = 1$ for every such l , while $(2/p) = 1$, contradicting $(h/p) = -1$. If $\beta = c^2 > 0$, use $p\beta \equiv 1 \pmod{h}$ instead. The shapes $\alpha = 0$ or $\beta = 0$ were already excluded by $v \neq R, D$. $\square$

4. EVERY DIAGONAL EXTERIOR SOURCE HAS AN ACTUAL MIDDLE RETURN

Theorem 4.1 (Cofactor collision crosses the channels). *Suppose an original exterior state at $p \equiv 1 \pmod{8}$ has $R = D$. Set*

$$z = (R - 1)/2 = \delta t^2,$$

where δ is the positive squarefree part (odd exponents retained). Then $\delta \equiv 3 \pmod{4}$, t is odd,

$$\delta \mid a, \quad \delta \mid p + 1, \quad 3 \leq \delta < R.$$

With $c = (p + 1)/\delta$ and $j = (\delta + 1)/4$, the explicit original middle state is

$$(h_M, r_M, s_M, \lambda_M) = (j, 1, c, 1), \quad a_M = jc, \quad R_M = c + 1, \quad u_M = j, \quad Q_M = \delta. \quad (10)$$

It returns (a_M, pa_M, pj) , including the original complementary middle orientation. Its small cofactor satisfies $\min(R_M, Q_M) < R$. There is also the original endpoint exterior state

$$R_0 = \delta, \quad a_0 = u_0 = (p + \delta)/4, \quad (h_0, r_0, s_0, \kappa_0) = (a_0, 1, 1, c), \quad (11)$$

with $a_0 < a$.

Proof. Since $R^2 = 4u + 1$, u is even, hence $a = hrs$ is even. Thus $R \equiv 7 \pmod{8}$ and $z \equiv 3 \pmod{4}$. The equality $u = z(z + 1)$ shows that every prime factor of z divides a . Consequently its squarefree part δ divides a , and $p = 4a - R \equiv -1 \pmod{\delta}$. Also $z = \delta t^2$ gives t odd and $\delta \equiv 3 \pmod{4}$.

Now $p = \delta c - 1 = 4jc - c - 1$. Thus (10) has $R_M = 4a_M - p = c + 1$, $r_M + s_M = R_M \lambda_M$, and $u_M \mid a_M^2$. The first-half upper bound is $c(\delta - 1) > 2$, true because $c \geq 2$ and $\delta \geq 3$. All denominators are positive and multiplication verifies the identity. Its second cofactor is exactly $\delta \leq (R - 1)/2$. The endpoint in (11) follows by the same direct substitution; $\delta < R$ gives $a_0 < a$. The endpoint formulas themselves are elementary antecedents; the new inference here extracts their needed divisor from the *actual* diagonal source rather than assuming its existence. $\square$

The map is not claimed injective after its original source is forgotten. Its complete fibre above a marked (10) retains the positive odd integer t , with

$$R = 2\delta t^2 + 1 < p, \quad a = (p + R)/4, \quad u = \delta t^2(\delta t^2 + 1) \mid a^2. \quad (12)$$

These explicit conditions and the squarefree marking are sufficient and necessary. They reconstruct the source by gcd normalization. The finite bound on its fibre is $t^2 \leq (p - 3)/(2\delta)$. Source aggregation sums these integer fibres; it does not replace them by a normalized count of residue classes.

The original residual itself need not decrease under the middle return. At $p = 2129$, the exterior source $(a, u, R, D) = (570, 5700, 151, 151)$ has $z = 75 = 3 \cdot 5^2$. Its middle return has $Q_M = 3$ but

$R_M = 711$. The strict measure in the theorem is the smaller cofactor, not R_M alone. At the hard prime $p = 48049$, $(a, u, R, D) = (12090, 24180, 311, 311)$ gives $\delta = 155, t = 1, j = 39, c = 310$ and the actual middle identity

$$\boxed{\frac{4}{48049} = \frac{1}{12090} + \frac{1}{580912410} + \frac{1}{1873911}}. \quad (13)$$

The old E divisor 24180 is not an M divisor at that shell; the return uses 39.

5. COMPLETE HARD-PRIME RIGIDITY AT SHAPE RADIUS SEVEN

Theorem 5.1 (Small-shape rigidity). *For a hard prime and an original E state with*

$$\max(|v - R|, |v - D|) \leq 7,$$

there is an even integer $n \geq 2$ such that

$$\boxed{h = 4n^2 - 2, \quad r = n, \quad s = 1, \quad R = D = 4n^2 - 1, \quad \kappa = 4n^2 - n - 1,} \quad (14)$$

$$p = 16n^3 - 4n^2 - 8n + 1, \quad a = hn, \quad u = hn^2.$$

At the same original shell it forces the two middle states

$$\boxed{u_M = n, \quad u'_M = nh^2, \quad (h_M, r_M, s_M, \lambda_M) = (n, 1, h, 1)} \quad (15)$$

and the complementary orientation has

$$(h'_M, r'_M, s'_M, \lambda'_M) = (n, h, 1, 1), \quad Q'_M = 4a - 1.$$

The first new ordered denominator triple is (nh, pnh, pn) , and $Q_M = 4n - 1 < R$.

Proof. There are 52 ordered coefficient profiles with $\alpha, \beta \in [-7, 7] \setminus \{0\}$ and $\alpha \equiv \beta \pmod{4}$: the four residue groups have sizes 2, 4, 4, 4. Their complete $B = \alpha\beta - 1$ spectrum, written as value followed by multiplicity, is

$$\begin{array}{c|cccccccccccccccccccc} B & -37 & -36 & -17 & -16 & -13 & -8 & -5 & -4 & 0 & 3 & 4 & 8 & 11 & 15 & 20 & 24 & 35 & 48 \\ \# & 2 & 4 & 2 & 4 & 4 & 4 & 2 & 4 & 2 & 2 & 4 & 2 & 4 & 2 & 4 & 2 & 2 & 2. \end{array}$$

If $B \neq 0$, direct multiplication of these small integers gives the following complete list of profiles for which $|B|$ has a prime factor greater than seven:

B	(α, β)	h forced by $h \mid B, (h/p) = -1$	
11	$(2, 6), (6, 2), (-2, -6), (-6, -2)$	11	(16)
-13	$(-6, 2), (2, -6), (-2, 6), (6, -2)$	13	
-17	$(-4, 4), (4, -4)$	17	
-37	$(-6, 6), (6, -6)$	37.	

Every other nonzero B is supported on 2, 3, 5, 7, contrary to Lemma 3.1. No unbounded factor search is implicit in (16). Substituting $v = hs^2$ and $Y = 2hrs$ into (4) gives

$$\boxed{4r^2 = hs^4 - (\alpha + \beta)s^2 + B/h}. \quad (17)$$

For $h = 11$, $\alpha \equiv 2 \pmod{4}$ would require $11s^2 \equiv 1 \pmod{4}$ from $R \equiv 3$, impossible. For $h = 17$, $\alpha \equiv 0$ would require $17s^2 \equiv 3 \pmod{4}$, also impossible.

For $h = 13$, the same residual condition forces s odd. If $\alpha + \beta = 4$, (17) is $4r^2 \equiv 8 \pmod{16}$, impossible. If $\alpha + \beta = -4$, it gives r even. Then $v \equiv 5 \pmod{8}$ and $\alpha \equiv 2 \pmod{8}$, so $R \equiv 3$ and $p = 4hrs - R \equiv 5 \pmod{8}$, again impossible.

For $h = 37$, s is odd and (17) makes r odd. The condition $p \equiv 1 \pmod{8}$ selects $\alpha = -6, \beta = 6$. Modulo three, (17) excludes $3 \mid s$ and forces $3 \mid r$. But then $p = 4hrs - 37s^2 - 6 \equiv 2 \pmod{3}$, contrary to the hard class. This excludes every nonsingular profile.

When $B = 0$, the positive-square lemma excludes $(1, 1)$, leaving $(-1, -1)$. Here $v = R - 1 = D - 1$, and $RD - 1 = 4hr^2$ becomes

$$s^2(v + 2) = 4r^2.$$

Coprimality implies $s \mid 2$. The choice $s = 2$ makes r odd and would give $v + 2 = r^2$ with $v = 4h$, impossible modulo four. Hence $s = 1$, $h = v = 4r^2 - 2$, which proves (14). Reduction of p modulo eight forces $r = n$ even.

For the same-shell return, $n + a = n(h + 1) = nR$. Thus $u_M = n \mid a^2$ is an original middle hit. Gcd normalization gives (15) and its complement. These are two distinct oriented middle states, not one E state with a new label. $\square$

The map from the singular E subfamily to the M orientation $r_M = 1$ is bijective onto the subfamily in (15): read $n = h_M$, then $h = 4n^2 - 2$ and every original coordinate in (14). The two M orientations retain one additional Boolean. The corresponding endpoint E return has the particularly simple formulas

$$R_0 = 4n - 1, \quad a_0 = n(h - n + 1), \quad u_0 = a_0, \quad \kappa_0 = h.$$

Here $a - a_0 = n(n - 1) > 0$. Thus the asymptotically sharp family in the previous exterior cutoff was already reducible to an explicit middle state and a strictly smaller endpoint E state. This does not lower that predecessor's bound on *all individual states*; it gives an existence-preserving way to remove this branch from a search for genuinely E-only inputs.

The hard restriction is necessary. At $p = 17$, the original E state $(a, u, h, r, s, \kappa) = (5, 5, 5, 1, 1, 6)$ has $\alpha = 2, \beta = -2$, so its nonsingular shape has radius two.

6. THE FIRST POSSIBLE NONSINGULAR SHAPE AT RADIUS EIGHT

Theorem 6.1 (Exact next-profile reduction). *At a hard prime, any nonsingular original E state with shape radius at most eight must have*

$$\boxed{(\alpha, \beta, h) = (8, -4, 11), \quad 4r^2 = 11s^4 - 4s^2 - 3, \quad p = 44rs - 11s^2 + 8.} \quad (18)$$

Here r, s are odd and coprime. Necessary hard-class restrictions include

$$rs \equiv -1 \pmod{3}, \quad r^2 \equiv s^2 \equiv 1 \pmod{5}, \quad rs \equiv 1 \pmod{5}.$$

Proof. Only profiles containing 8 or -8 are new beyond (16); their other coordinate is in $\{-8, -4, 4, 8\}$. Any positive coordinate 4 is excluded by Lemma 3.1. The products of absolute value 64 give $B = 63$ or -65 . The first has only small prime factors. In the second, the only possible nonresidue grades are 13 and 65, both $1 \pmod{4}$, whereas $\alpha \equiv 0$ and $R \equiv 3$ would require $hs^2 \equiv 3 \pmod{4}$.

The pair $(-8, -4)$, in either order, has $B = 31$ and forces $h = 31, s$ odd. Equation (17) gives $4r^2 \equiv 12 \pmod{16}$, impossible. For the remaining $B = -33$ pairs the only possible grades are 11 and 33. The grade 33 is excluded modulo four. At grade 11, equation (17) and the prime congruence modulo eight leave only $(8, -4)$; it makes r, s odd and gives (18). For completeness, the twelve new ordered pairs and every grade divisor containing a prime greater than seven give exactly the following sixteen rows:

(α, β)	B	h
$(-8, -4), (-4, -8)$	31	31
$(-8, 4), (4, -8)$	-33	11, 33
$(-8, 8), (8, -8)$	-65	13, 65
$(-4, 8), (8, -4)$	-33	11, 33
$(4, 8), (8, 4)$	31	31.

The omitted pairs $(-8, -8)$ and $(8, 8)$ have $B = 63$, supported only on 3 and 7. Thus no admissible radius-eight grade row is hidden by the table. This calculation can equivalently be exhausted over $r, s \pmod{24}$ using (17) modulo 48, as in the certificate.

Modulo three, $3 \mid s$ would force $3 \mid r$, violating coprimality. Thus r, s are nonzero modulo three, and $p \equiv 1$ gives $rs \equiv -1$. Modulo five, (18) allows only $s^2 = r^2 = 1$. Then $p \equiv 2 - rs \pmod{5}$; the hard residue condition $p \equiv 1, 4 \pmod{5}$ forces $rs \equiv 1$. Modulo seven, (18) gives

$$r^2 \equiv s^4 - s^2 + 1 \pmod{7}.$$

The square classes show that either $s \equiv 0$, $r^2 \equiv 1$, $p \equiv 1$, or $s^2 = r^2 = 1$; in the latter case the hard residue condition forces $rs \equiv -1$ and $p \equiv 2 \pmod{7}$. Together with $p \equiv 1 \pmod{5}$ and $p \equiv 1 \pmod{24}$ this sharpens the possible hard classes to

$$p \bmod 840 \in \{1, 121\}.$$

□

Corollary 6.2 (Radius eight already has an original middle coefficient). *Every hard-prime E state of shape radius at most eight forces an original M state. In the singular case this is (15). In the sole possible nonsingular case (18), the exact return is*

$$\boxed{s_M = (p+3)/11, \quad (h_M, r_M, s_M, \lambda_M) = (1, 3, (p+3)/11, 1),}$$

$$\boxed{a_M = 3(p+3)/11, \quad u_M = 9, \quad R_M = (p+36)/11, Q_M = 11, \quad (x, y, z) = (a_M, ps_M, 3p).}$$
(18a)

It attains the preceding hard-prime middle localization equality $11a_M = 3(p+3)$.

Proof. The grade identity (6) gives $p \equiv \alpha = 8 \pmod{11}$, so s_M is integral. Then $p = 11s_M - 3 = 4 \cdot 3s_M - (3 + s_M)$. A common factor of 3 and s_M would divide the hard prime p , impossible. The first-half inequalities follow because $p \equiv 1 \pmod{8}$ gives $s_M \equiv 4 \pmod{8}$, hence $s_M > 3$. Substitution proves both the original M gate and its ordered identity. The original E coordinates remain in the graph of this map; if they are forgotten, its fibre is the set of positive coprime quartic pairs in (18) with the fixed value of p . Theorem 2.1 bounds that fibre by three. No integral-point classification is needed to prove this return. □

Equation (18) remains a specific integral-point problem for classifying the exterior records, but no longer an obstruction to returning an original middle state from any such record. The bounded scan in the certificate checks odd $s \leq 200000$ exactly. No absence beyond that range, effective integral-point cutoff, or global ES conclusion is inferred. The small point $r = s = 1$ gives $p = 41$, outside the hard classes. The elementary Pell rewriting

$$(11s^2 - 2)^2 - 11(2r)^2 = 37$$

also retains the extra condition that $(X+2)/11$ be a square; dropping that condition would change the problem.

7. A NATURAL CUBIC INVOLUTION FAILS THE REQUIRED FIXED-PRIME RETURN

For $B \neq 0$, (4) has the explicitly checked rational involution

$$\Psi(X, Y) = (B/X, BY/X^2), \quad X \neq 0. \quad (19)$$

Substitution proves preservation of (4), and applying it twice is the identity. For an original positive state it can have positive X' only when $B > 0$. Integrality of X' requires

$$v \mid B \iff v \mid 4u \iff s^2 \mid 4r^2 \iff s \mid 2.$$

If $s = 2$, then r is odd, $X' \equiv 3 \pmod{4}$, and $Y' = rX'$ is odd, so the inverse $a' = Y'/2$ fails integrality. If $s = 1$, the complete coordinate transport is

$$\begin{aligned} h' &= v' = 4r^2 + h - R - D, & r' &= r, & s' &= 1, \\ R' &= 4r^2 - D, & D' &= 4r^2 - R, \\ a' &= rh', & u' &= h'r^2, & \kappa' &= D' - r, \\ p' &= 4rh' - R'. \end{aligned} \quad (20)$$

Whenever its signs are positive, it has $R', D' \equiv 1 \pmod{4}$ and $p' \equiv 3 \pmod{4}$, not the original hard prime. More exactly,

$$\boxed{p' - p = (4r - 1)(4r^2 - R - D).} \quad (21)$$

The second factor is $2 \pmod{4}$, hence cannot vanish. Thus (19) is an actual rational correspondence, but not a same- p selector. At $p = 2129$, its positive composite-parameter image has $p' = 5951 =$

11 · 541, $R' = D' = 249$, $h' = 155$, $r' = 10$, $s' = 1$. This records what the map does rather than treating its failure as evidence that the two geometries are unrelated.

The obvious reverse attempt from the always available residual-one source in the class $p' \equiv 3 \pmod{4}$ also has an exact obstruction. Write $p' = 4hrs - 1$, $R = 1$, $D = 4hr^2 + 1$, $v = hs^2$ with $\gcd(r, s) = 1$. For (19),

$$X' = 4r^2/s^2 + hs^2 - 4hr^2 - 2.$$

An integer image still requires $s \mid 2$. At $s = 1$, $X' = h - 2 - 4(h - 1)r^2 < 0$. At $s = 2$, $X' = 4h - 2 - (4h - 1)r^2 \leq -1$. For $s > 2$ the image is not integral. Thus the complete residual-one source does not supply a positive integral inverse for the hard-prime problem. This does not prohibit other positive $p' \equiv 3 \pmod{4}$ parameter records; the displayed image at 5951 is one such composite record. It identifies why the immediate reverse construction is insufficient.

8. CONTINUATION STATUS AND EXECUTABLE BOUNDARY

The legitimate new reductions are: the diagonal source has a same-prime middle return with smaller cofactor; all hard-prime radius-seven E states are singular and already have M hits at the same shell; the next radius has the unique nonsingular quartic (18), whose grade congruence itself forces (18a). The marked curve includes the original fixed- p line (8), which its natural involution does not preserve.

None of this proves that an arbitrary prescribed prime supplies an original E or M state. The universal Turn 7 existence milestone remains outstanding. The finite profile classification is exhaustive because its domain consists of the 52 explicitly enumerated integer profiles; the bound on the quartic's search prefix is a different, finite verification scope.

The supplied main verifier enumerates all original E square divisors at every $p \equiv 1 \pmod{8}$ prime through its declared bound. The separate implementation uses primitive (h, r, s) enumeration and a different modular profile test. Both keep the original integers and compare source fibres rather than comparing only state totals. All check outcomes, failures and time limits are recorded by the execution runner. No independent human mathematical review or Lean build is claimed.

REFERENCES

- [1] Elsholtz, C., & Tao, T. (2013). *Counting the number of solutions to the Erdős–Straus equation on unit fractions*. Journal of the Australian Mathematical Society, 94, 50–105. [arXiv:1107.1010](#), §2, Propositions 2.2 and 2.6.
- [2] Hatcher, A. (2022). *Topology of Numbers*. Author edition, §6.4, pp. 205–212. Quadratic reciprocity and supplementary laws.
- [3] The Clankers. (2026). *Turn 7 continuation: exterior three-parameter bound and complete original atlas*. Supplied source `exterior_atlas/core.tex`, sections 2–5, within `ES_Turns_6_7_Validated_Handoff_20260919.zip`. The complete source is preserved, not represented as independently reviewed.